

CO 1 ASSESMENT TOOL 3
FLOW CHART ALGORITHM

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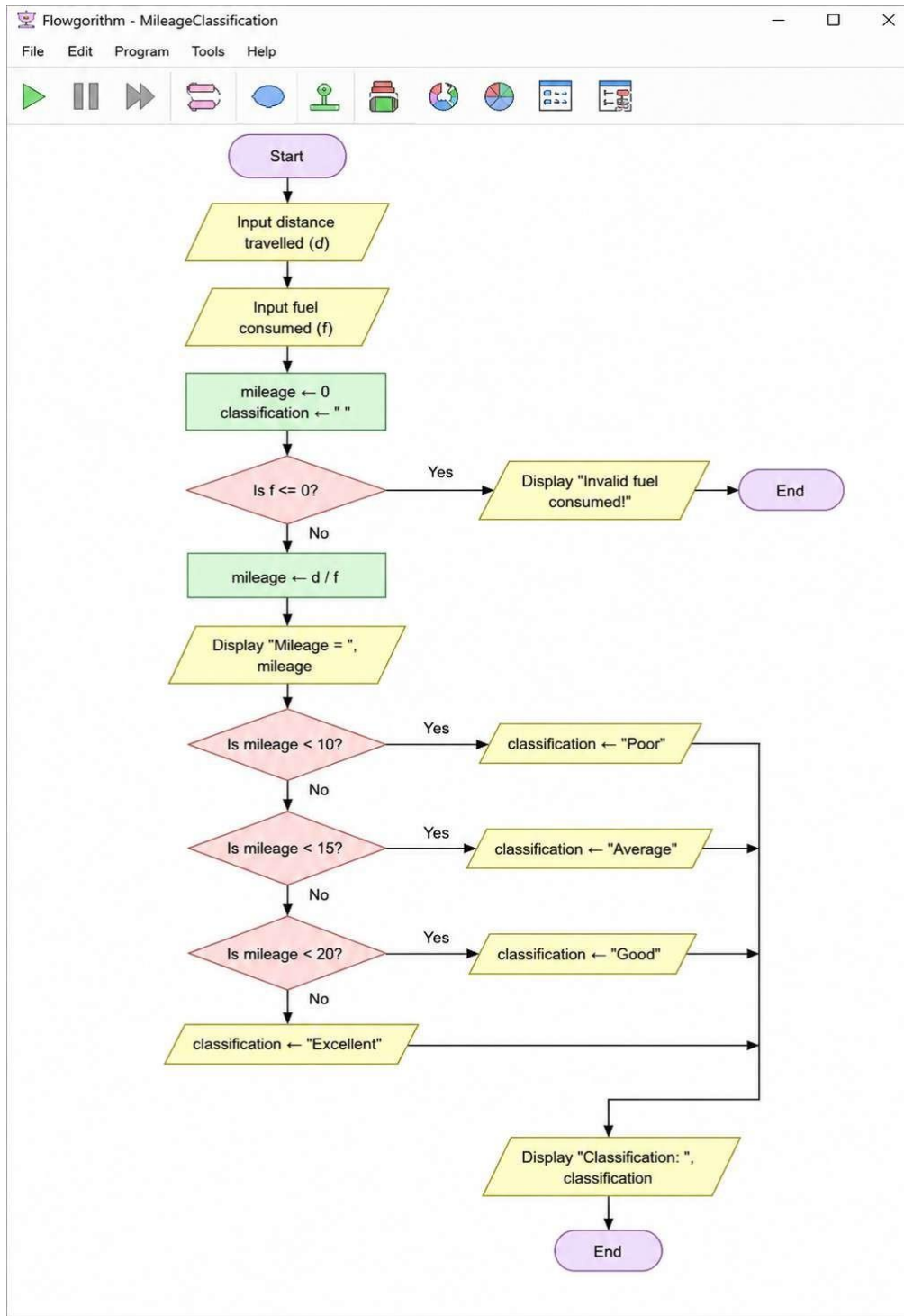
REG.NO : 192465066

COURSE CODE : CSA0924

COURSE NAME : JAVA PROGRAMMING

SLOT : D

1. Read distance travelled and fuel consumed. Calculate mileage and classify it as Poor, Average, Good, or Excellent.



PESUDO CODE:

START

Declare Distance, Fuel, Mileage as Real

Input Distance

Input Fuel

IF Fuel $\leq$ 0 THEN

 Display "Invalid fuel consumed."

ELSE

 Mileage = Distance / Fuel

 Display "Mileage = ", Mileage

 IF Mileage < 10 THEN

 Display "Poor"

 ELSE IF Mileage < 15 THEN

 Display "Average"

 ELSE IF Mileage < 20 THEN

 Display "Good"

 ELSE

 Display "Excellent"

 END IF

END IF

STOP

CODE:

```
import java.util.Scanner;

public class MileageClassification {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        double distance, fuel, mileage;

        System.out.print("Enter distance travelled: ");

        distance = sc.nextDouble();

        System.out.print("Enter fuel consumed: ");

        fuel = sc.nextDouble();

        if (fuel <= 0) {

            System.out.println("Invalid fuel consumed!");

        } else {

            mileage = distance / fuel;

            System.out.printf("Mileage = %.2f%n", mileage);

            if (mileage < 10) {

                System.out.println("Classification: Poor");

            } else if (mileage < 15) {

                System.out.println("Classification: Average");

            } else if (mileage < 20) {

                System.out.println("Classification: Good");

            } else {

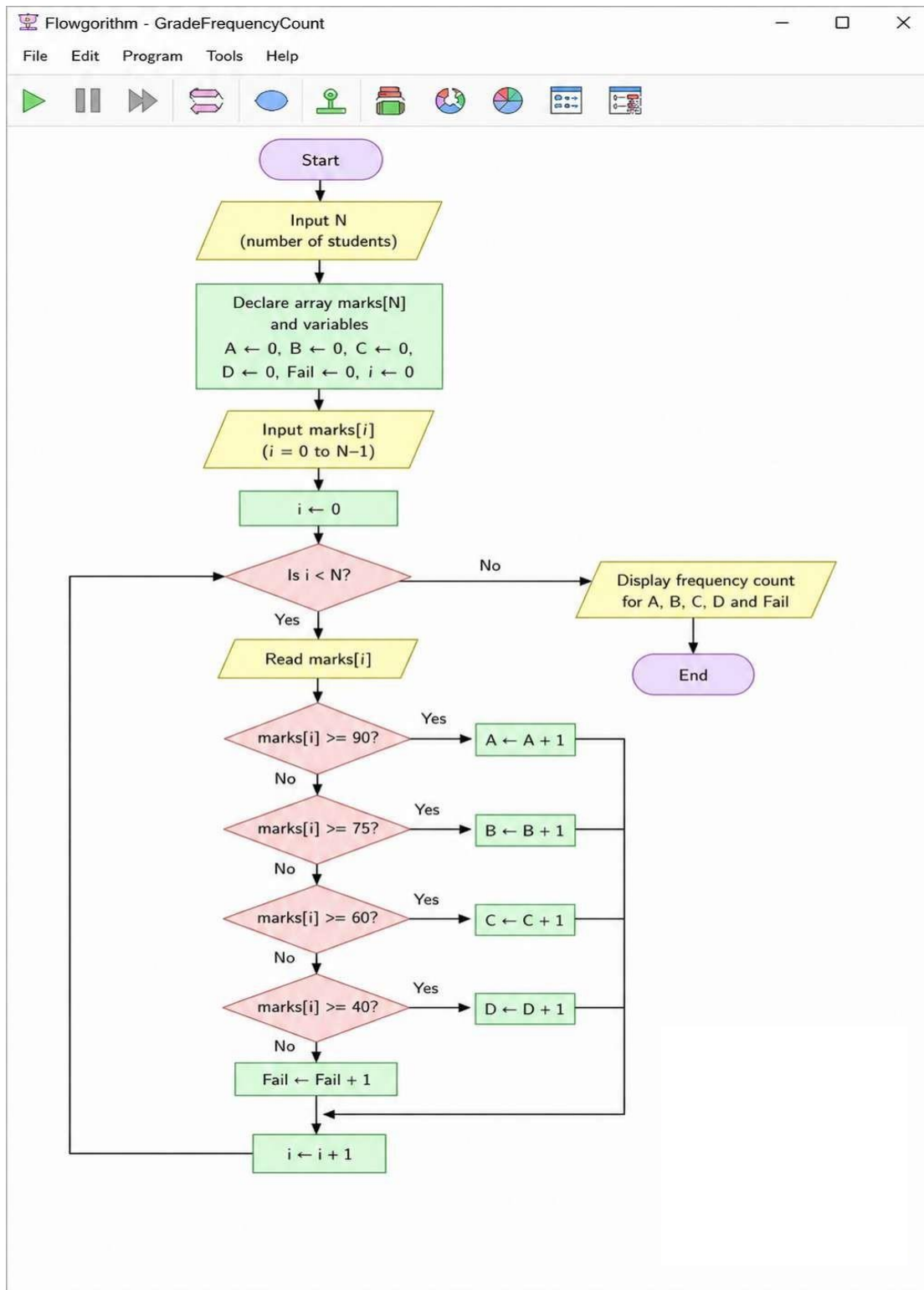
                System.out.println("Classification: Excellent");

            }

        }

        sc.close();    }}
```

2. Read N marks into an array and display a frequency count for grade categories A, B, C, D, and Fail.



PESUDO CODE:

START

DECLARE N, i : INTEGER

DECLARE marks[N] : INTEGER ARRAY

DECLARE A, B, C, D, Fail : INTEGER

READ N

A $\leftarrow$ 0

B $\leftarrow$ 0

C $\leftarrow$ 0

D $\leftarrow$ 0

Fail $\leftarrow$ 0

FOR i $\leftarrow$ 0 TO N - 1 DO

 READ marks[i]

 IF marks[i] $\geq$ 90 THEN

 A $\leftarrow$ A + 1

 ELSE IF marks[i] $\geq$ 75 THEN

 B $\leftarrow$ B + 1

 ELSE IF marks[i] $\geq$ 60 THEN

 C $\leftarrow$ C + 1

 ELSE IF marks[i] $\geq$ 40 THEN

 D $\leftarrow$ D + 1

 ELSE

Fail $\leftarrow$ Fail + 1

END IF

END FOR

DISPLAY "Grade Frequency Count"

DISPLAY "A Grade : ", A

DISPLAY "B Grade : ", B

DISPLAY "C Grade : ", C

DISPLAY "D Grade : ", D

DISPLAY "Fail : ", Fail

STOP

JAVA CODE:

```
import java.util.Scanner;
```

```
public class GradeFrequencyCount {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the number of students: ");
```

```
        int N = sc.nextInt();
```

```
        int[] marks = new int[N];
```

```
        int A = 0;
```

```
        int B = 0;
```

```
        int C = 0;
```

```
        int D = 0;
```

```
        int Fail = 0;
```

```
        for (int i = 0; i < N; i++) {
```

```
            System.out.print("Enter marks of student " + (i + 1) + ": ");
```

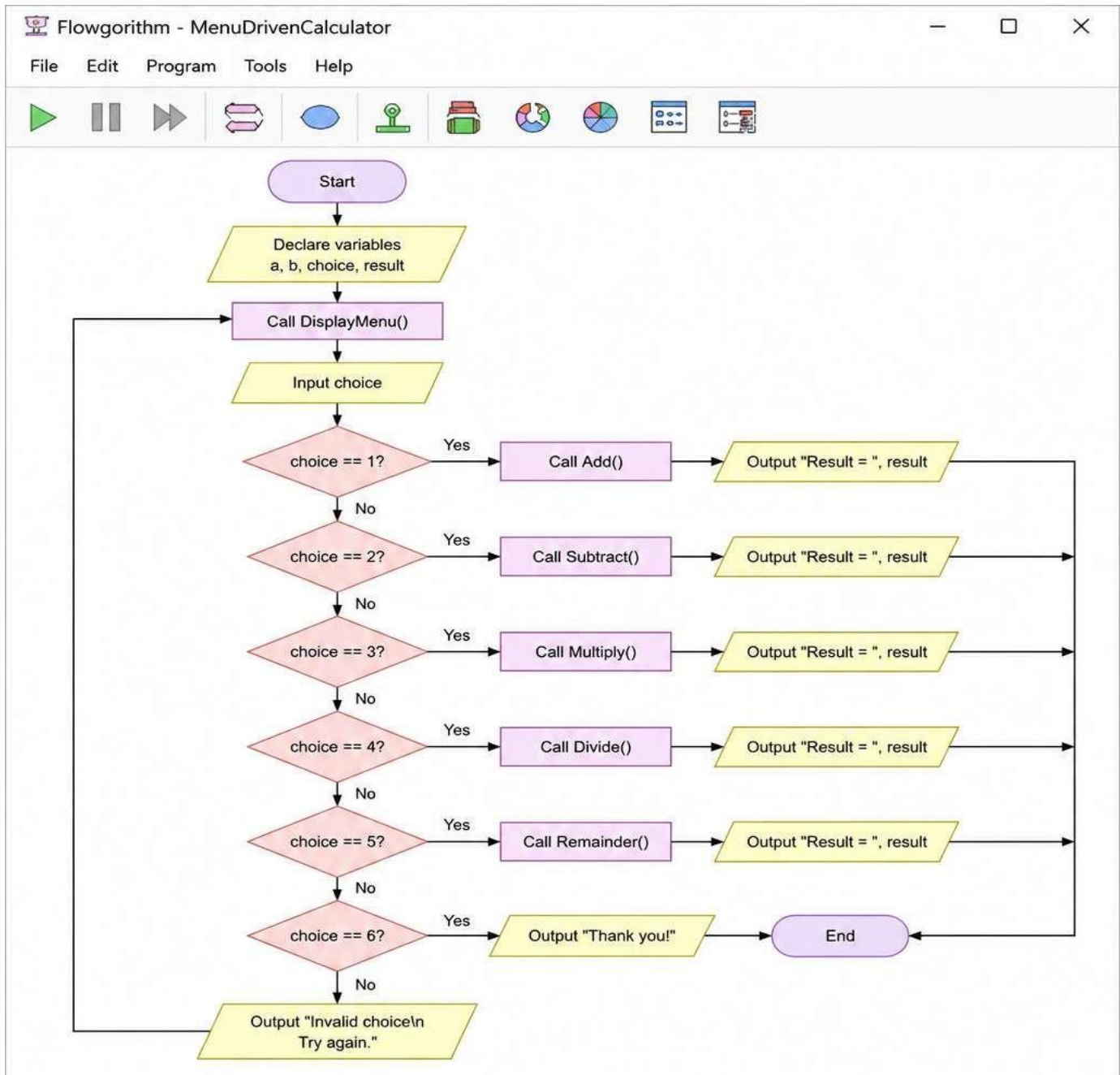
```

marks[i] = sc.nextInt();
if (marks[i] >= 90) {
    A++;
}
else if (marks[i] >= 75) {
    B++;
}
else if (marks[i] >= 60) {
    C++;
}
else if (marks[i] >= 40) {
    D++;
}
else {
    Fail++;
}
}

System.out.println("\nGrade Frequency Count");
System.out.println("----- ");
System.out.println("A Grade : " + A);
System.out.println("B Grade : " + B);
System.out.println("C Grade : " + C);
System.out.println("D Grade : " + D);
System.out.println("Fail : " + Fail);
sc.close();
}
}

```

3. Create a menu-driven calculator using separate functions for addition, subtraction, multiplication, division, and remainder.



PESUDO CODE

START

FUNCTION Add(a, b)

 RETURN a + b

END FUNCTION

FUNCTION Subtract(a, b)

 RETURN a - b

END FUNCTION

FUNCTION Multiply(a, b)

 RETURN a * b

END FUNCTION

FUNCTION Divide(a, b)

 IF b = 0 THEN

 DISPLAY "Cannot divide by zero!"

 RETURN 0

 ELSE

 RETURN a / b

 END IF

END FUNCTION

FUNCTION Remainder(a, b)

 IF b = 0 THEN

 DISPLAY "Cannot find remainder by zero!"

 RETURN 0

 ELSE

 RETURN a % b

 END IF

```
END FUNCTION

DECLARE choice : INTEGER
DECLARE a, b, result : REAL

DISPLAY "==== MENU ====="
DISPLAY "1. Addition"
DISPLAY "2. Subtraction"
DISPLAY "3. Multiplication"
DISPLAY "4. Division"
DISPLAY "5. Remainder"
DISPLAY "6. Exit"

READ choice

IF choice >= 1 AND choice <= 5 THEN
    READ a
    READ b

    END IF

    SWITCH(choice)

CASE 1
    result ← Add(a, b)
    DISPLAY "Result = ", result
    BREAK

CASE 2
    result ← Subtract(a, b)
    DISPLAY "Result = ", result
    BREAK

CASE 3
```

```
result ← Multiply(a, b)
DISPLAY "Result = ", result
BREAK
```

CASE 4

```
result ← Divide(a, b)
DISPLAY "Result = ", result
BREAK
```

CASE 5

```
result ← Remainder(a, b)
DISPLAY "Result = ", result
BREAK
```

CASE 6

```
DISPLAY "Thank You!"
BREAK
```

DEFAULT

```
DISPLAY "Invalid Choice"
```

END SWITCH

STOP

JAVA CODE:

```
import java.util.Scanner;

public class MenuDrivenCalculator {

    static double add(double a, double b) {
        return a + b;
    }

    static double subtract(double a, double b) {
```

```
        return a - b;
    }

    static double multiply(double a, double b) {
        return a * b;
    }

    static double divide(double a, double b) {
        if (b == 0) {
            System.out.println("Cannot divide by zero!");
            return 0;
        }
        return a / b;
    }

    static double remainder(double a, double b) {
        if (b == 0) {
            System.out.println("Cannot find remainder by zero!");
            return 0;
        }
        return a % b;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        double a, b, result;

        System.out.println("===== MENU =====");
        System.out.println("1. Addition");
        System.out.println("2. Subtraction");
        System.out.println("3. Multiplication");
```

```
System.out.println("4. Division");
System.out.println("5. Remainder");
System.out.println("6. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch (choice) {
    case 1:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
        result = add(a, b);
        System.out.println("Result = " + result);
        break;
    case 2:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
        result = subtract(a, b);
        System.out.println("Result = " + result);
        break;
    case 3:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
```

```
        result = multiply(a, b);

        System.out.println("Result = " + result);

        break;
    case 4:

        System.out.print("Enter first number: ");

        a = sc.nextDouble();

        System.out.print("Enter second number: ");

        b = sc.nextDouble();

        result = divide(a, b);

        System.out.println("Result = " + result);

        break;
    case 5:

        System.out.print("Enter first number: ");

        a = sc.nextDouble();

        System.out.print("Enter second number: ");

        b = sc.nextDouble();

        result = remainder(a, b);

        System.out.println("Result = " + result);

        break;
    case 6:

        System.out.println("Thank You!");

        break;
    default:

        System.out.println("Invalid Choice!");

    }

    sc.close();
}
```